

An exact null for two Rasch protocols

The reliable change index and the PCAR/t-test of unidimensionality

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Two protocols, one statistic

Two widely used procedures compare a pair of Rasch person estimates with a pooled standard error, then refer the ratio to 1.96.

- **Individual change** over time, in `RMpersonChange()`
- **Unidimensionality**, the RUMM2030 PCA/t-test protocol (Smith Jr. 2002; Tennant and Pallant 2006)

Both read 1.96 as the 5% two-sided normal cutoff. Neither null is normal on a short scale, and both errors run in the same direction.

The Rasch change index

For a person measured on two occasions,

$$RCI = \frac{\hat{\theta}_2 - \hat{\theta}_1}{\sqrt{SE(\hat{\theta}_1)^2 + SE(\hat{\theta}_2)^2}}$$

`RMpersonChange()` offers `critical = "simulate"` rather than 1.96, because the reference is off on short scales. The question this work answers is what the correct reference is, and whether it needs simulating at all.

Why the null is not $N(0, 1)$

Self-normalization. The two standard errors are deterministic functions of the realized scores, so the denominator is not independent of the numerator. A move to an extreme score inflates numerator and denominator together, and the ratio is damped. Both tails are pulled in, so the cutoff falls **below** 1.96.

Discreteness. With complete data the sum score is sufficient, so $\hat{\theta}$ takes only as many values as there are scores. The null is discrete, not continuous.

A t reference does not rescue it: a t denominator is independent of its numerator, which is the one property this denominator lacks.

An exact null, no simulation

Sufficiency makes the null tractable.

- $\hat{\theta}(r)$ and $SE(r)$ are deterministic lookups, the ones `RMscoreSE()` builds
- $P(r \mid \theta)$ comes from the Lord–Wingersky recursion, the elementary symmetric function identity in probability form
- Under the null the occasions are independent given θ , so the joint is a discrete distribution over $(k + 1)^2$ score pairs

Nothing is simulated. Sufficiency fixes the **value** in each cell, Lord–Wingersky supplies the **weight**. That is what “exact” means here.

How was the critical value derived?

There is no formula. The critical value is **counted**, not derived.

1. A short scale has few possible total scores, so few possible pairs of before and after scores: 361 for six items with four categories
2. Each pair has a fixed change index and a known probability under no change
3. Sort the pairs by change index and read off where the most extreme 2.5% begins at each end

The answer is the same every time, and it agrees with thousands of simulated datasets. Large moves towards the ends of the scale come with large standard errors, which is why the value lands below 1.96.

The cutoff tracks test length

Exact upper cutoff, four response categories throughout:

Items	4	6	10	20	$\rightarrow \infty$
Cutoff	1.64	1.73	1.82	1.87	1.96
Reliability	.78	.83	.89	.94	$\rightarrow 1$

Monotone in length, converging on 1.96. Referring the RCI to a normal null is therefore **conservative**, and increasingly so the shorter the scale. The gap from 1.96 is largest where reliability is already respectable, so an acceptable reliability is no assurance that the cutoff has converged.

Targeting, not skew

Sample skew barely moves the symmetry of the pooled cutoff. Targeting does.

- Pushing a sample 2.2 logits off target, so ~40% land at an extreme score, drops the cutoff from 1.70 to 1.52
- The pooled cutoff is a property of the **sample** as well as the items

Recalibrating on each sample holds the level even when targeting is poor. Carrying a value across samples does not: a cutoff calibrated on a well-targeted sample gave a realized size of .028 against a nominal .05 when applied off target, where recalibration held .042.

The PCA/t-test unidimensionality protocol is the change index in disguise.

The PCA/t-test protocol

The protocol (Smith Jr. 2002; Tennant and Pallant 2006; Hagell 2014):

1. Principal component analysis of the Rasch residuals
2. Split items by loadings above +0.3 and below -0.3
3. Estimate a person location from each subset
4. Per-respondent t-test of the two locations against 1.96
5. Refer the proportion significant to a 5% benchmark

Hagell (Hagell 2014) showed the verdict depends on sample size and on the binomial interval method, and read it as ordinary hypothesis testing.

It is the change index with subtests in place of occasions

$$t = \frac{\hat{\theta}_A - \hat{\theta}_B}{\sqrt{SE(\hat{\theta}_A)^2 + SE(\hat{\theta}_B)^2}}$$

The same two mechanisms carry over, so the reference is again below 1.96. One difference matters: the effective test length is the **subtest**, not the scale. A 20-item scale split 10/10 is governed by two 10-item instruments, and the protocol is often run near the recommended floor of 12 thresholds (Tennant and Pallant 2006; Horton and Tennant 2010).

The 1.96 rule is not a 5% test

Exact size of the per-respondent rule, across everything tried:

- Runs from **1.4% to 3.9%**, never reaching 5%
- Too lenient by a factor of **1.3 to 3.7**
 - biased toward declaring unidimensionality
- Worst on dichotomous items and off-target samples
- **44 to 46%** of respondents cannot be flagged at all: no attainable pair of subtest scores clears 1.96, so their contribution is zero with probability one

At 40 items split 20/20 the rule is still short by a factor of 1.27.

A mechanism for Hagell's finding

Comparing an observed proportion against a benchmark that is too lenient shifts the whole power curve toward passing.

- Conclusions flipping with sample size is ordinary power behaviour against a biased reference, not a pathology of the binomial interval
- The direction is uniform, so the protocol is **biased toward declaring unidimensionality**

The fix is cheap: the enumeration that states the correct reference is the same recursion and lookup the package already builds.

One threat the change null does not cover

Both protocols assume conditional independence. For the change index that is an auxiliary assumption, not the hypothesis. Carry-over, a time-2 answer depending on the time-1 answer beyond θ (Marais 2009), breaks it.

With cutoffs calibrated under independence the test stays **conservative**, but the false positives polarise and a flat treatment effect reads as a severity gradient:

d	Type I (20 items)	flat 0.5 logit effect, above – below
0	.049	–0.00
0.5	.027	+0.20
1.0	.013	+0.33

Racking both occasions into one analysis and screening the same-item pairs with `RMlocdepQ3Cutoff()` detects it well below the level that does damage.

Response dependence is state dependence

Within-person dependence across occasions has two sources, and the repeated-measures literature keeps them apart.

Source	Rasch term	Longitudinal / panel term
a shared latent trait	person location θ	random intercept, the ICC
the realized past answer moves the next	response dependence	state dependence (Heckman 1981)

The first is expected and is what makes the change interpretable. The second is a violation. Marais shifts the time-2 difficulty by $(1 - 2x_1)d$, a lagged realized response, which is state dependence rather than heterogeneity. Mainstream models place it in the residual covariance, a serial or AR structure, or in a transition model, not in the random effects (Diggle et al. 2002).

Two connections the Rasch change literature does not draw:

Takeaways

- The RCI and the PCA/t-test share one statistic, and 1.96 is the wrong cutoff for both
- The correct null is discrete and enumerable, no simulation
- Both errors point one way: change is **under-detected**, multidimensionality is **under-detected**
- The cutoff is a property of the sample, so recalibrate rather than reuse

What is next

- A parametric bootstrap of the whole PCA/t-test pipeline, including the data-chosen split, to settle whether selection or the biased reference dominates. The exact null above serves as its reference
- The RCI orders respondents by standardised distance from the null, not by how much they changed. A conditional critical **change** in logits comes from the same enumeration at no extra cost

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